

Part B Question 1

Algorithm:- Run a loop for n times for each guest take input season, room type and No. of nights. Calculate the price by given information. If No. of nights is greater than 7 then apply a discount of 15% on the total price and output total price. Calculate the Total Revenue and output it.

P A C:- ~~Chart~~

Given Data	Processing	Required Results
- Season, room type - No. of nights	- Calculate rate depends on season and room type - Calculate total price by multiplying rate with nights - Check for nights if it exceeds then apply discount	- Final Price - Total Revenue

IPO Chart:-

Input	Processing	OUTPUT
- Season - Room type - No. of Nights	- Rate = Season rate Evaluate Season and Room type and calculate the rate - Total Price = Rate * No. of Nights - Discount = Total Price * 15% - Reduce total price if No. of Nights > 7	- Final price - Total Revenue

Flow chart:-



Pseudocode:

$i \leftarrow 0$

Total Revenue $\leftarrow 0$

Price $\leftarrow 0$

Rate $\leftarrow 0$

WHILE $i \leq N$ DO,

INPUT Season, R t, Nights

IF Season == "Peak" THEN

Case OF R t

"Standard" : ~~Price~~ Rate $\leftarrow 5000$

"Deluxe" : Rate $\leftarrow 8000$

"Suite" : Rate $\leftarrow 12000$

ELSE

Case OF R t

"Standard" : Rate $\leftarrow 3000$

"Deluxe" : Rate $\leftarrow 5000$

"Suite" : Rate $\leftarrow 8000$

END IF

Price $\leftarrow$ Rate * Nights

IF Nights > 7 THEN

Price $\leftarrow$ Price * 0.85

ENDIF

OUTPUT Price

Total Revenue $\leftarrow$ Price + Total Revenue

$i \leftarrow i + 1$

END WHILE

OUTPUT Total Revenue